

Integration — Lesson 32 Test: Partial Fractions

10 questions · show your work in the box · no calculator · exact answers ($\ln 2$, $\ln(3/2)$) · verify every split by re-adding the pieces

Name: _____ Date: _____ Score: _____ / 10

1. Use the log rules to write each as a single logarithm:

1 pt

- (a) $\ln|x - 5| - \ln|x + 2|$ (b) $\ln 12 - \ln 4$ (c) $3 \ln 2$ (d) $2 \ln|x - 1| + \ln|x + 4|$

2. Decompose only (do NOT integrate). Find A and B in

1 pt

$7/((x - 3)(x + 4)) = A/(x - 3) + B/(x + 4)$, then verify by adding your two pieces back together.

3. Decompose only. Find A and B in $3x/((x - 4)(x + 2)) = A/(x - 4) + B/(x + 2)$,^{1 pt} and verify by re-adding. (The top is not just a number — evaluate it at each magic value too.)

4. Find $\int \frac{4}{(x-1)(x+3)} dx$. Show the split, verify it by re-adding, then integrate. 1 pt

5. Find $\int \frac{2}{x(x+2)} dx$. (What is the magic value for the bare factor x?) 1 pt

6. Look hard at this one BEFORE you set up any A's and B's: find 1 pt

$\int \frac{(2x+3)}{(x^2+3x-10)} dx$. State which tool you used and explain in one sentence why partial fractions would be the slow road here.

7. Find $\int \frac{(3x+1)}{(x^2-x-6)} dx$. Factor the bottom first, and show your f'/f check before committing to partial fractions. 1 pt

8. Compute $\int_0^1 \frac{2}{(x+1)(x+3)} dx$ exactly, and squeeze your answer into a single logarithm.

9. Find $\int \frac{5x}{(x-1)(x+4)} dx$, then use the log rules to write your answer as one single logarithm. 1 pt

